VAPO: Value-Model-Based RL for Advanced Reasoning — 60.4 on AIME 2024

ByteDance Seed

Base model: Qwen2.5-32B (no SFT data)

Headline result: 60.4 on AIME 2024 (avg@32)

Outperforms DAPO (50) and DeepSeek-R1-Zero-Qwen-32B (47) by 10+ points

Why Value Models? Finer Credit Assignment for Long Chain-of-Thought Reasoning

Value-model-free (GRPO, DAPO)

- Trajectory-level final reward
- Uniform advantage to all tokens
- Higher variance Monte Carlo signal
- Coarse credit assignment in long CoT

Value-model-based (VAPO)

- Token-level credit assignment
- Lower-variance per-token estimates
- Generalizes value across samples
- Better for subtle multi-step errors

Three Challenges Plaguing Value-Model-Based RL in Long-CoT Tasks

1) Value model bias

Long trajectories + bootstrapping amplify estimation bias.

2) Heterogeneous lengths

Different response lengths need different bias-variance tradeoffs.

3) Reward sparsity

Only final <eos> reward in long CoT weakens learning signal.

VAPO Framework: Integrated Solutions for Value-Model-Based Reasoning RL

Challenges	VAPO Components
Value model bias	Value-Pretraining; Decoupled-GAE
Heterogeneous lengths	Length-adaptive GAE
Reward sparsity	Positive Example LM Loss; Group-Sampling
Policy optimization stability	Clip-Higher; Token-level Loss

Length-Adaptive GAE: Dynamic λ Adjustment for Variable Response Lengths

Short responses

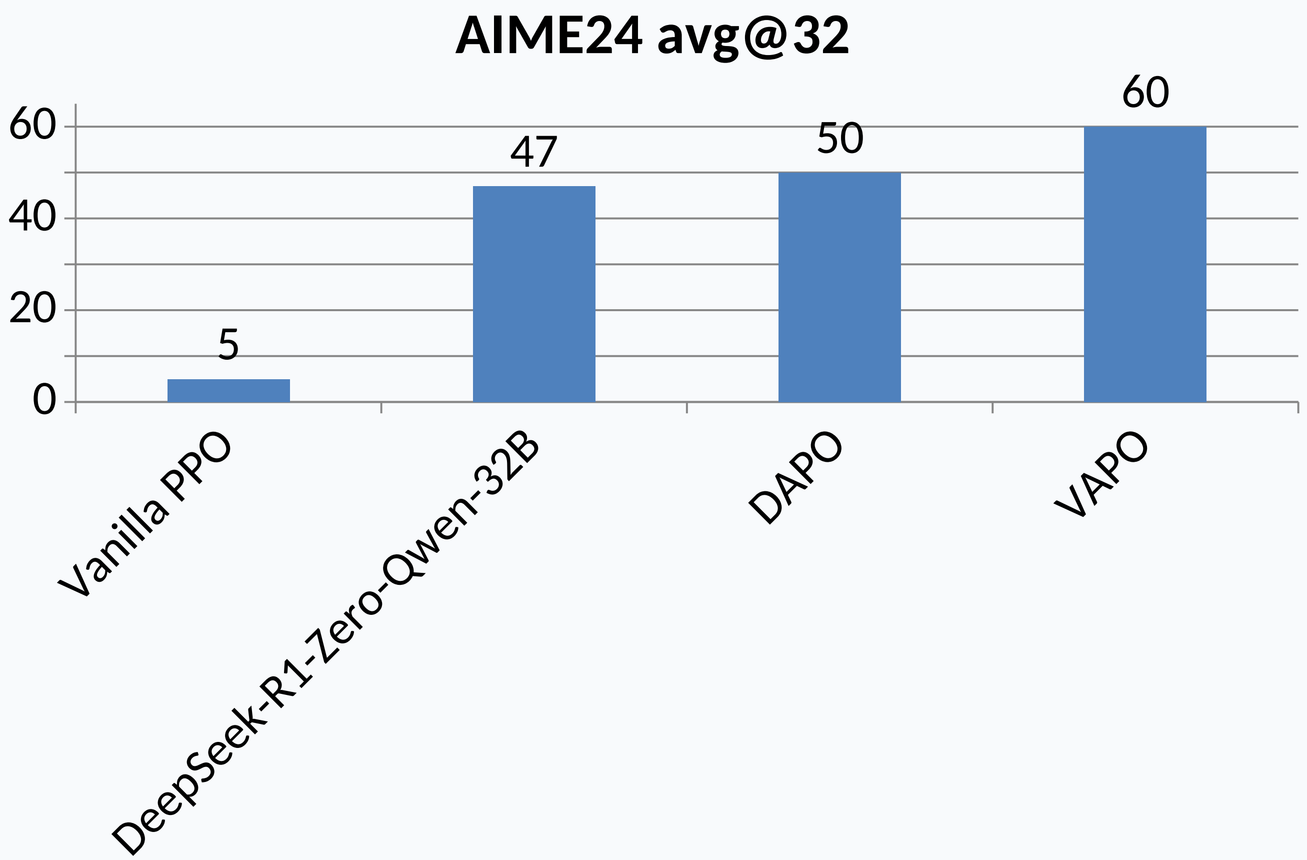
Lower λ → lower variance

Long responses

Higher λ → lower bias

Length-adaptive GAE dynamically tunes λ by response length, optimizing the bias-variance tradeoff across heterogeneous long-CoT trajectories.

VAPO Scores 60 on AIME 2024 — 10+ Points Above DAPO and DeepSeek-R1-Zero



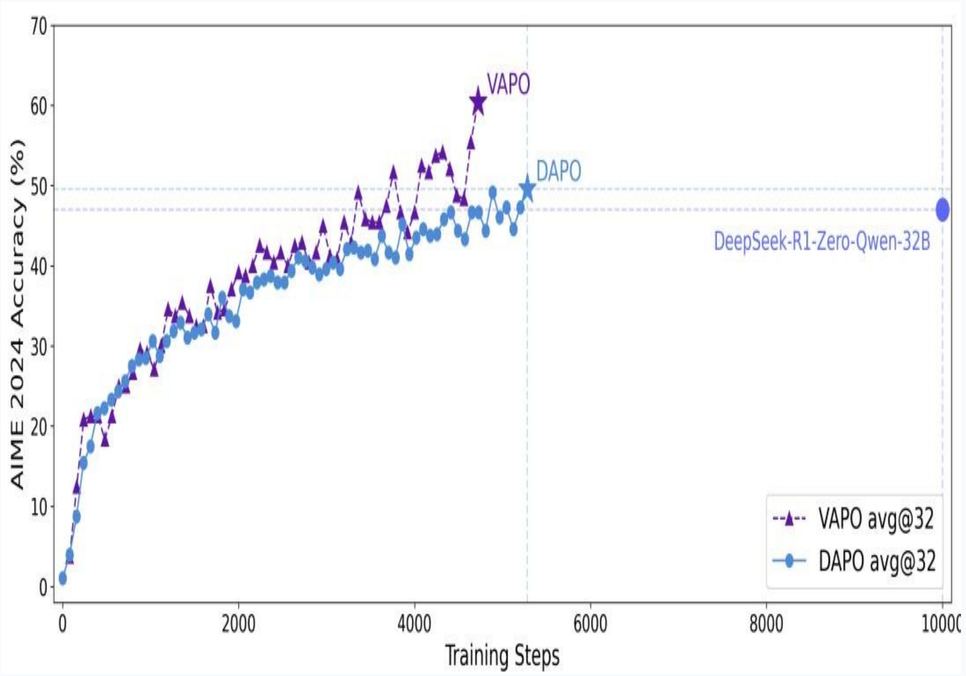
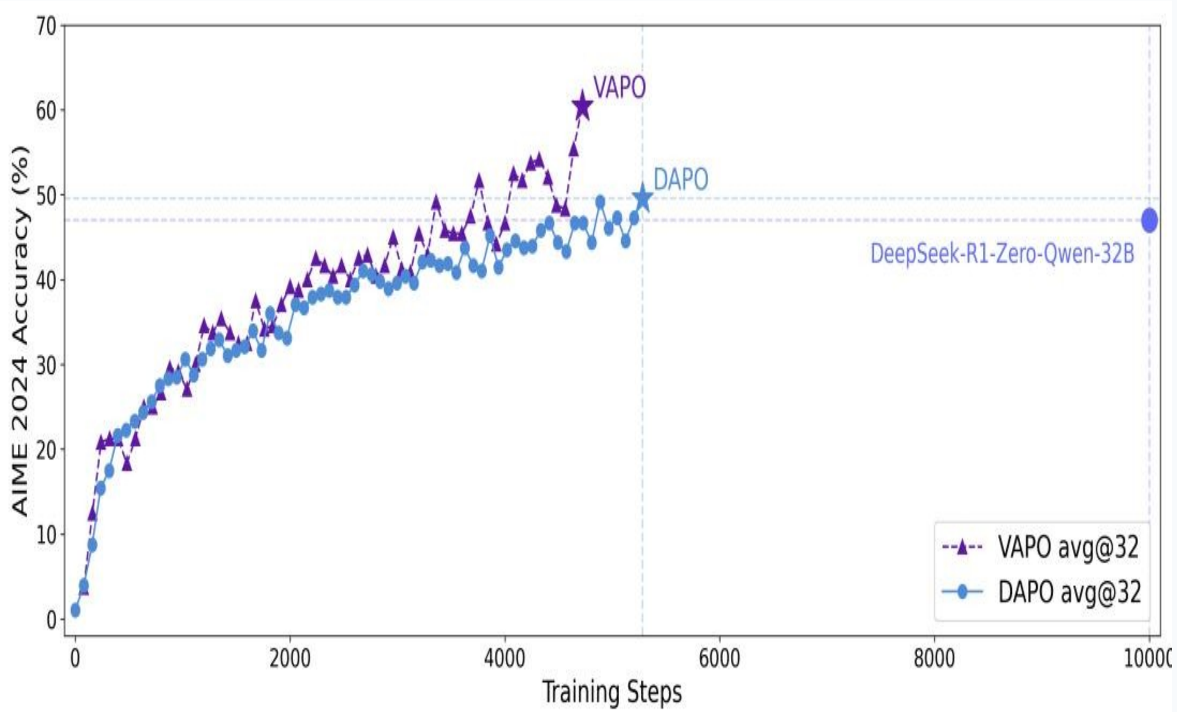
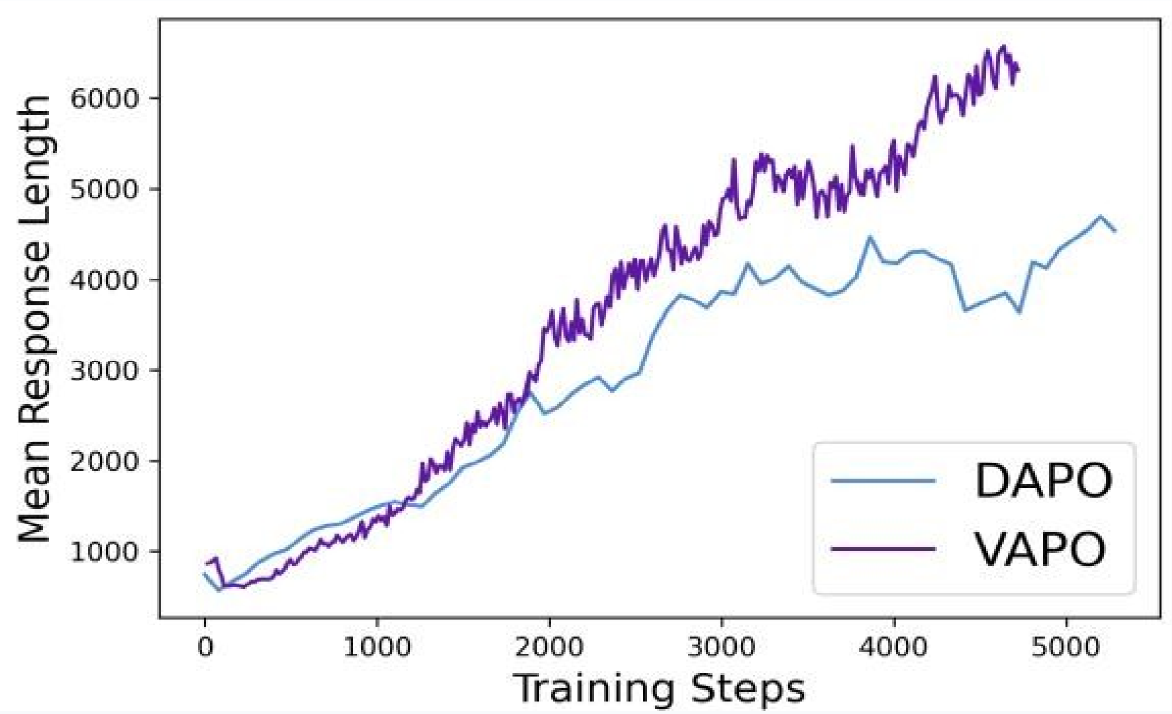


Figure 2: VAPO consistently outperforms DAPO throughout training.

VAPO Reaches SOTA in 5K Steps — Half the Training Budget of DAPO





- VAPO reaches ~60% around step 4,500; DAPO plateaus near 50% around step 5,500.
- VAPO response length grows to ~6,000 tokens vs DAPO ~4,500, indicating more elaborate reasoning chains.

Ablation: Every Component Matters — Value-Pretraining Alone Adds 49 Points

Configuration	AIME24
VAPO (full)	**60**
w/o Value-Pretraining	11
w/o Decoupled-GAE	33
w/o Length-adaptive GAE	45
w/o Clip-Higher	46
w/o Token-level Loss	53
w/o Positive Example LM Loss	54
w/o Group-Sampling	55

Largest drops from full VAPO (60):

- No Value-Pretraining: -49
- No Decoupled-GAE: -27
- No Length-adaptive GAE: -15

Key Takeaways: Value-Model-Based RL Is Back — With the Right Engineering

- Value-model-based RL can outperform value-model-free methods when core failure modes are engineered away.
- VAPO achieves 60.4 on AIME 2024, exceeding DAPO (50) and DeepSeek-R1-Zero-Qwen-32B (47) by 10+ points.
- Value-Pretraining is the most critical component (11 → 60 with all components).
- Length-adaptive GAE is essential for heterogeneous response lengths in long-CoT reasoning.
- VAPO shows stable, efficient training: SOTA in ~5K steps and no crashes across multiple runs.